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= 2 × (relative atomic mass of hydrogen atom)

= 2 × 1.008 u

= 2.016 u.

ii) Relative molecular mass of glucose ($C_6H_{12}O_6$)

= (6 × 12) + (12 ×1.008) + (6 ×16)

= 72+12.096+96

= 180.096 u

Table 1.1 Relative atomic masses of some elements

Element	Relative atomic mass	Element	Relative atomic mass
H	1.008	Cl	35.45
C	12	K	39.10
N	14	Ca	40.08
O	16	Cr	51.9
Na	23	Mn	54.94
Mg	24.3	Fe	55.85
S	32.07	Cu	63.55

Evaluate Yourself

2) Calculate the molar mass of the following.

(i) Ethanol(C_2H_5OH)

(ii) Potassium permanganate ($KMnO_4$)

(iii) Potassium dichromate ($K_2Cr_2O_7$)

(iv) Sucrose ($C_{12}H_{22}O_{11}$)

1.4 Mole Concept

Often we use special names to express the quantity of individual items for our convenience. For example, a dozen roses means 12 roses and one quire paper means 25 single sheets. We can extend this analogy to understand the concept of mole that is used for quantifying atoms and molecules in chemistry. Mole is the SI unit to represent a specific amount of a substance.

To understand the mole concept, let's calculate the total number of atoms present in 12 g of carbon -12 isotope or molecules in 158.03 g of potassium permanganate and 294.18 g of potassium dichromate.

12 Numbers = 1 Dozen



12 Roses



12 Balls



12 Apples

1 Mole ≡ 6.023×10^{23} entities



158.03g of $KMnO_4$



294.18g of $K_2Cr_2O_7$

Fig. 1.2 Mole Concept